

Phoenix Active Directory Home lab Investigation Report

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Findings

Time: 2026-07-22 02:50:00 UTC – 2026-07-22 04:30:00 UTC

Host: DC01 (172.16.50.10), Attacker IP (172.16.50.1)

Compromised Account: Bob (PHOENIX.LOCAL\Bob)

Indicators of Compromise:

- **Execution Method:** Remote service creation via Impacket PsExec
- **Privilege Level Achieved:** NT AUTHORITY\SYSTEM
- **Location of Activity:**
 - C:\Windows\system32 (SYSTEM shell)
 - C:\Windows\Temp (registry hive dumps)
- **Artifacts Created:** Uploaded executable, registry hive dump files
- **Persistence Mechanism:** Creation of a Domain Administrator account named Bobby.

Investigation

On 2026-07-22 at approximately 03:00:00 UTC, a new account known as Bobby in a Domain Controller was created without knowledge of anyone in the Phoenix Active Directory home lab. An Investigation was Immediately conducted to identify the reason for the account creation.

At 02:51:13 UTC, Kerberos enumeration revealed valid domain accounts. The attacker attempted AS-REP roasting and Kerberoasting, but both techniques failed due to domain hardening and enforced authentication requirements.

At 03:05:40 UTC, the attacker pivoted to password spraying using the previously enumerated usernames.

At 03:15:00 UTC, the password spray succeeded, compromising the Bob account.

At 03:22:36 UTC, the attacker attempted to authenticate into Bob and was successful, and telemetry confirmed that Bob possessed local administrator privileges on the Domain Controller.

At 03:22:24 UTC, the attacker used a remote login service to a service in order to access the SYSTEM account.

At 03:25:00 UTC, The attacker successfully uploaded a payload to the ADMIN\$ share, and executed remote code. This resulted in a full NT AUTHORITY\SYSTEM shell on DC01.

At 04:22:47 UTC, the attacker created an account called **Bobby**, and at 04:23:19 UTC, the account was added as a Domain Administrator.

At 04:26:41 UTC the attacker cleared Windows event logs, based on evidence, it is unclear what logs were cleared.

Additionally, the attacker also cleared the Application, Security, and System logs to hide their presence.

No persistence mechanisms were created, and no additional hosts were targeted. The activity concluded at 04:30:00 UTC.

WHO

- **Host:** DC01 (172.16.50.10)
- **Attacker:** Unknown (simulated adversary in homelab)
- **Accounts Involved:**
 - Bob (compromised)
 - SYSTEM (privilege escalation)
 - Bobby (malicious persistence account)

WHAT

A threat actor enumerated domain accounts, attempted Kerberoasting and AS-REP roasting, pivoted to password spraying, compromised Bob's account, escalated to SYSTEM via PsExec, dumped registry hives, created a new Domain Administrator account, and cleared event logs to hide activity.

WHEN

- **02:51:13 UTC:** Kerberos enumeration
- **02:51–03:05 UTC:** AS-REP roasting & Kerberoasting attempts (failed)
- **03:05:40 UTC:** Password spraying begins
- **03:15:00 UTC:** Bob account compromised
- **03:22:24–03:22:36 UTC:** SYSTEM access obtained
- **03:25:00 UTC:** PsExec remote execution
- **04:22:47 UTC:** Bobby account created
- **04:23:19 UTC:** Bobby added to Domain Admins
- **04:26:41 UTC:** Event logs cleared
- **04:30:00 UTC:** Activity concluded

WHERE

Activity occurred on:

- **Host:** DC01
- **Directories:**
 - C:\Windows\system32
 - C:\Windows\Temp
 - ADMIN\$ share

WHY

The compromise occurred due to:

- Weak password hygiene enabling password spraying
- Bob having unintended **local administrator** privileges on DC01
- SMB remote service creation being enabled
- Lack of endpoint detection for PsExec activity
- Insufficient monitoring of privileged account creation
- No alerting on event log clearing

The attacker's intent was likely domain dominance and long-term persistence.

HOW

- Kerberos enumeration
- Failed AS-REP and Kerberoasting attempts
- Password spraying succeeded
- Bob account compromised
- Remote login service used to obtain SYSTEM
- PsExec uploaded payload and executed remote code
- Registry hives dumped
- Malicious Domain Admin account created
- Event logs cleared to hide activity

Recommendations:

- 1) Immediately isolate the DC01 host and remove the malicious artifacts, including the uploaded executable and any registry hive dump files. Remove the unauthorized Bobby account and revoke its Domain Administrator privileges.
- 2) Reset the credentials for the Bob account and review authentication logs for additional suspicious logon attempts. Run Splunk queries for the attacker's IP address and remote service creation events to identify any other hosts exhibiting similar behavior. If additional hosts are found, isolate them immediately.
- 3) Although attacker IP addresses can be changed easily, consider blocklisting the attacker's known IPs and tightening firewall rules to prevent remote service creation and unauthorized administrative access. Additionally, implement monitoring for privileged account creation to prevent similar compromises.

Screenshots:

2026-07-22 02:51:13.574	4768	WIN-PC01\$	1
2026-07-22 02:51:13.584	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:51:13.637	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:51:13.638	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:51:13.760	4768	WIN-PC01\$	1
2026-07-22 02:51:13.763	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:51:13.924	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:51:14.107	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:51:14.108	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:51:21.410	4768	WIN-PC01\$	1
2026-07-22 02:51:21.411	4769	WIN-PC01\$@PHOENIX.LOCAL	1
2026-07-22 02:56:15.349	4768	vIindsay	1
2026-07-22 02:56:15.353	4768	vIindsay	1
2026-07-22 03:01:33.707	4768	vIindsay	1

Figure 1: Enumeration of Kerberos services

New Search

index="winevent" EventCode= 4769

table	_time	Account_Name	Client_Address	Service_Name	Ticket_Options	Failure_Code
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|sort - _time|

Time range: Date time range

✓ 45 events (7/21/26 2:50:00.000 AM to 7/22/26 4:30:00.000 AM) No Event Sampling

Events (45) Patterns Statistics (45) Visualization

Show: 20 Per Page Format Preview: On

_time	Account_Name	Client_Address	Service_Name	Ticket_Options	Failure_Code
2026-07-22 04:12:16.285	DC01\$@PHOENIX.LOCAL	::1	DC01\$	0x40810000	0x0
2026-07-22 04:12:16.278	DC01\$@PHOENIX.LOCAL	::1	DC01\$	0x40810000	0x0
2026-07-22 04:10:46.604	vIindsay@PHOENIX.LOCAL	::1	DC01\$	0x40810000	0x0
2026-07-22 03:56:24.947	vIindsay@PHOENIX.LOCAL	::1	DC01\$	0x40810000	0x0
2026-07-22 03:04:35.192	vIindsay@PHOENIX.LOCAL	::1	DC01\$	0x40810000	0x0
2026-07-22 02:51:21.411	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	WIN-PC01\$	0x40810000	0x0
2026-07-22 02:51:14.108	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	krbtgt	0x60810010	0x0
2026-07-22 02:51:14.107	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	DC01\$	0x40800000	0x0
2026-07-22 02:51:13.924	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	WIN-PC01\$	0x40810000	0x0
2026-07-22 02:51:13.763	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	DC01\$	0x40800000	0x0
2026-07-22 02:51:13.638	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	krbtgt	0x60810010	0x0
2026-07-22 02:51:13.637	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	DC01\$	0x40810000	0x0
2026-07-22 02:51:13.584	WIN-PC01\$@PHOENIX.LOCAL	::ffff:172.16.50.40	DC01\$	0x40800000	0x0

Figure 2: Kerberos Successful TGT

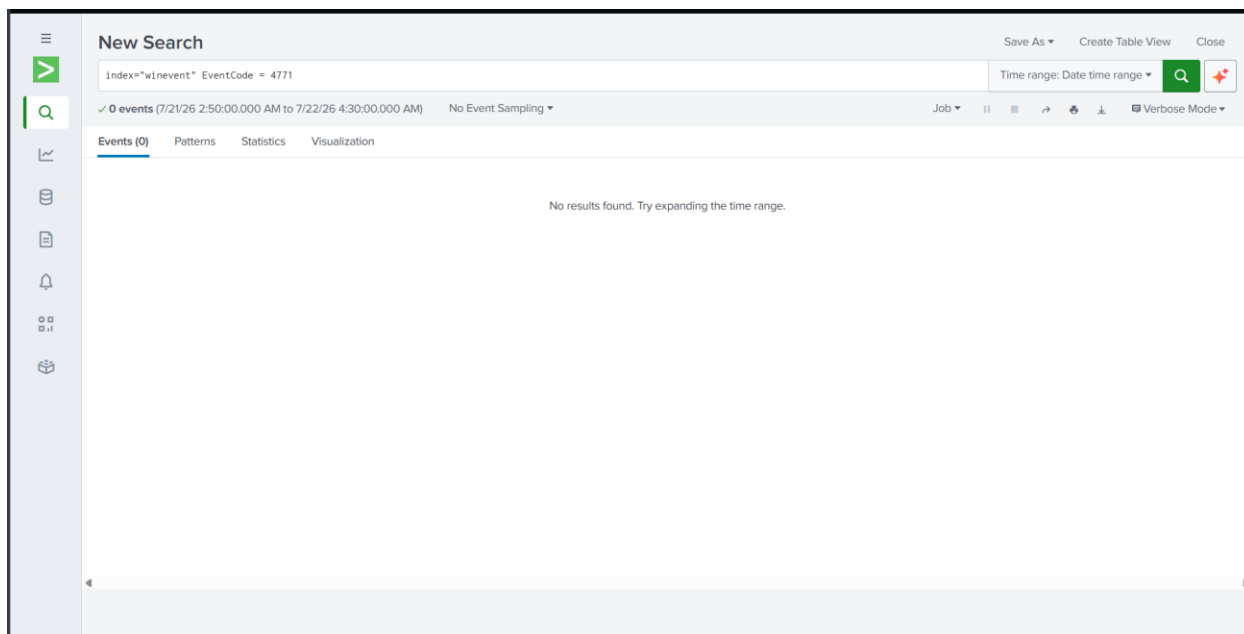


Figure 3: Checking for failed reauthentication requests

New Search

Save As Create Table View Close

index=*winevent* EventCode=4625
| stats count by _time, Source_Network_Address, Account_Name, Failure_Reason
| sort _time Time range: during Jul 2026

✓ 29 events (7/1/26 12:00:00.000 AM to 8/1/26 12:00:00.000 AM) No Event Sampling

Events (29) Patterns Statistics (52) Visualization

Show: 20 Per Page Format Preview: On

_time	Source_Network_Address	Account_Name	Failure_Reason	count
2026-07-22 03:05:40.598	172.16.50.1	-	Unknown user name or bad password.	1
2026-07-22 03:05:40.598	172.16.50.1	Bob	Unknown user name or bad password.	1
2026-07-22 03:05:42.688	172.16.50.1	-	Unknown user name or bad password.	1
2026-07-22 03:05:42.688	172.16.50.1	Bob	Unknown user name or bad password.	1
2026-07-22 03:08:24.643	172.16.50.1	-	Unknown user name or bad password.	1
2026-07-22 03:08:24.643	172.16.50.1	vlindsay	Unknown user name or bad password.	1
2026-07-22 03:09:41.170	172.16.50.1	-	Unknown user name or bad password.	1
2026-07-22 03:09:41.170	172.16.50.1	vlindsay	Unknown user name or bad password.	1
2026-07-22 03:11:07.342	172.16.50.1	-	Unknown user name or bad password.	1
2026-07-22 03:11:07.342	172.16.50.1	vlindsay	Unknown user name or bad password.	1
2026-07-22 03:14:44.916	172.16.50.1	-	Unknown user name or bad password.	1
2026-07-22 03:14:44.916	172.16.50.1	Bob	Unknown user name or bad password.	1

Figure 4: Failed logons – Password Spraying

Figure 7: Service Installation

splunk>Enterprise

Figure 8: Service names

